Robert Lee

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SUMMARY OF QUALIFICATIONS

- **Extensive coding experience:** 10+ years across C, C++, C#, Python, Java, TypeScript, MATLAB, Git, CMake.
- **Machine Learning & Computer Vision:** CNN projects in human pose estimation, semantic road segmentation, monocular depth estimation.
- **Embedded systems:** KiCad, VHDL, ARM Assembly, Altium. Designed efficient and fast low-level code.
- **Exceptional academics:** Recipient of 20+ prestigious academic & leadership awards, valued at $100,000+.
- **Proven leadership skills:** Lead organizer for 200+ conference. 6 years leading 180+ volunteers to reach 650+ seniors as Senior's Program Founder. Delivered workshops to 350+ engineering students as IEEE SB Chair.

WORK EXPERIENCE

Senior Software Engineer – Azure AI Search Mar. 2025 – present
Microsoft, *Redmond, WA (Remote)*

- Core contributor to vector search and search relevance, driving collaborations across multiple teams and delivering robust, high-quality code; proactively identified and addressed challenges; led design meetings.
- Provided careful and thorough code reviews, offering detailed and constructive feedback by leveraging my technical expertise in distributed systems and vector algorithms to improve system reliability and performance. Key contributor on major features critical to vector search, including multi-modal/multi-vector search.
- As a subject matter expert, coordinated cross-collaborative investigations and root-caused several deeply technical production incidents effectively, ensuring customer services are quickly restored to health and the underlying defects are understood and repair items triaged.
- Improving search relevance to ground GenAI, Agents, LLMs, Retrieval Augmented Generation & semantic search.

Software Engineer II – Azure AI Search Jun. 2022 – Feb. 2025
Microsoft, *Redmond, WA (Remote)*

- Delivered 8-32x cost savings and up to 20x latency reduction for customers by implementing vector quantization techniques such as binary vectors, scalar and binary quantization, with SIMD acceleration.
- Improved relevance stack by defining and designing hybrid search subscores and score thresholding.
- Implemented new quota enforcement mechanism for HNSW indexes underlying physical resource utilization, leveraging data-driven decisions and cross-team design discussions, reducing limit overshoot by 100x.
- Successfully managed the end-to-end development of improved facet aggregations features from a spec doc; implemented a novel and highly extensible lexer-parser-evaluator which leveraged advanced mathematical concepts such as BNF grammar, shunting yard, and Reverse Polish Notation to parse, simplify, and validate the grammar string; added extensive AB test coverage.
- Proactively identified test coverage weaknesses and expanded vector search engine tests with a new test suite, which discovered one critical bug in a new quantization algorithm before release.

Software Engineer – Azure AI Search Jul. 2021 – May 2022
Microsoft, *Redmond, WA (Remote)*

- Drove development work for a highly requested feature, revamped key component of telemetry pipeline.

Software Engineer Intern – Azure Cognitive Search Sep. 2020 – Dec. 2020
Microsoft, *Redmond, WA (Remote)*

- Designed & completed 2nd most requested API & backend feature on Azure Cognitive Search team in **C#**.
- Currently released under preview API.

Software Engineer Intern – Azure Search, AI Platform Jun. 2019 – Aug. 2019
Microsoft, *Bellevue, WA*

- Developed a dynamic search website generator with **suggestions** and **filtering** options in **TypeScript**.
- Connected designers, engineers, and program managers to identify scope of work and feature set.
- **Improved user experience** on Azure Search portal by adding new JSON editor & search website customization.

Software Developer Intern – Garage Program Jan. 2018 – Apr. 2018
Microsoft, *Vancouver, BC*

- Built cross-platform mobile app leveraging **offline machine learning** for chest x-ray classification in **C#**.
- Built **image processing** pipeline, DevOps Continuous Integration, **iOS share extension**, and integration of TensorFlow Android binding library. **Team expert on Git** version control.

EDUCATION

Bachelor of Electrical & Computer Engineering **University of Victoria,** Victoria, BC
Cumulative GPA: 97% Sept. 2016 – Apr. 2021

LEADERSHIP EXPERIENCE

Founder & Program Leader, Senior's Program Jul. 2015 – Oct. 2021

- Founded a series of workshops on technology and computers for seniors in the community.
- Supervised a team of **180 volunteers** to reach **650+ attendees** over 30 workshops; raised $700.

Chair & Vice-Chair, IEEE Student Branch Sep. 2019 – Jan. 2021

- Co-delivered **14** skill development workshops focusing on Git, machine learning, integrated circuits, breadboards, soldering, & circuit design, reaching **350+** engineering students; successfully secured $1000 in funding.

Conference Organizer Lead Aug. 2019 – Nov. 2019

- **Assembled** and orchestrated organizing committee; established conference vision.
- Spearheaded logistics planning for **200+ attendee conference** on "fusion of technology and business strategy".

PROJECTS

Human Pose Estimation using Deep Neural Networks github.com/robertklee/COCO-Human-Pose

- **Team Lead** to design a deep neural network for human pose estimation (HPE) on COCO-2017 dataset.
- Architected cloud training pipeline, model architecture, data augmentation, model visualization, and deployment.
- Our model achieves **very good performance** on most images. It struggles in some difficult images, typically with highly overlapped people or heavily occluded joints.

Computer Vision Project on Semantic Road Segmentation github.com/robertklee/KITTI-RoadSeg

- Successfully trained **U-Net CNN** on KITTI Road dataset with worst case of 91% F1 score using **Keras** & **Python**.
- Designed network architecture by analyzing numerous computer vision research articles.
- Designed data generator, train and test scripts, loss functions; configured cloud training; tuned hyperparameters.

C Optimization Project on Discrete Cosine Transform github.com/robertklee/C-Optimization-DCT

- Achieved a **10x speedup** compared to naïve implementation by using **C** and **assembly-level** optimizations.
- Configured **CMake** for platform agnostic compilation; profiled code using Valgrind; created custom asm operator.

Computer Vision Project on Monocular Depth Estimation github.com/DeclanMcIntosh/monodepthV2tf

- Successfully trained **U-Net CNN** for depth estimation on DrivingStereo dataset using **Keras** & **Python**.
- Constructed training loss (photometric reprojection and edge-aware smoothness) in Keras backend, which are designed to counter object occlusion and camera egomotion.

Battlesnake Reinforcement Learning-based AI Controller

- Challenge: Design algorithm to control snake in real-time game combat environment. Goal: Survive the longest.
- Trained keras-rl reinforcement learning model with a combination of self-play and publicly available snakes.

AWARDS AND ACHIEVEMENTS

Schulich Leader Scholarship 2016 – 2020

- **$80,000 full-ride scholarship**; 50 awarded nationally among 1512 nominees; selected for academic excellence in science, technology, engineering, and mathematics, and outstanding community or entrepreneurial leadership.

1st Place, Western Engineering Competition - Senior Design Jan. 2020

- Built a robot to collect Martian artifacts in a timed environment with limited budget and material.
- Qualified for **national** Canadian Engineering Competition.

Jamie Cassels Undergraduate Research Award for 2019-2020 Sept. 2019

- Under the mentorship of faculty supervisor, research **hardware acceleration** for ML neural networks.

1st Place (3 Times), UVEC Engineering Competition - Senior Design Oct. 2017, 2018, & 2019

- Developed the **best robotic solution** with limited materials in a timed environment.
- Represented university as a **delegate** at **Western Canada's** Western Engineering Competition.

3rd Place, Google Games Competition Oct. 2017

3rd Place, Engineering Design Autonomous Cable-Carrying Robot Project Mar. 2017

- Architected the robot's control program using Finite State Machine with basic control theory & signal processing.
- Challenge: The robot must find a target object within constrained search area and run a simulated cable from source to target while minimizing excess cable and object collisions.

1st Place, UVEC Engineering Competition - Junior Design Oct. 2016

- Identified the client's objectives, constraints; constructed the prototype; pitched the final product to judges.
- Served as delegate at regional **Western Engineering Competition** in Banff, Alberta.

Governor General's Academic Medal, Bronze 2016

National Champion, Michael Smith Science Challenge 2014

- Set **national record** score of 97.5% among 1,700+ candidates.